

The Capstone Challenge: Your AI Project

Welcome to your Capstone Challenge! This is the final, most important mission of our course. It's your opportunity to bring together everything you've learned—from AI concepts and prompting to creative design and architectural thinking—into a single, comprehensive project.

This isn't just another assignment. It's a blueprint for *a real solution to a real problem*. You will act as the founder, visionary, and chief architect for a new AI-powered application.

Objective

To use all your knowledge from this course to design a comprehensive, multi-page blueprint for an AI-powered application that solves a real-world problem you care about.

Your Task

You will act as an "App Architect." Your goal is to identify a problem in your community, for your family, or for India, and design a detailed blueprint for an app that helps solve it. **This is a design and strategy task—no coding is required!** You will be creating the master plan that a team of engineers and designers would use to build your vision.

How to Build Your Capstone Blueprint

Follow this structure to create a detailed, professional-grade project proposal.

The Foundation (The "Why" and "Who")

This page sets the stage. It explains what you're building and, most importantly, why it needs to exist.

Section 1: Project Name & The Problem

- **Project Name:** Give your app an inspiring and memorable name.
- **The Problem:** Clearly explain the problem you are solving. Who is affected? Why is this a problem? What are the current solutions, and why aren't they good enough? Be specific. (e.g., "Finding reliable caregivers for elderly parents is a stressful, time-consuming, and fragmented process for working professionals in urban India...")

Section 2: The Solution (Your Mission)

- Write a clear, one-paragraph "elevator pitch" for your solution. How does your app solve the problem you just defined? (e.g., "CareSync is a comprehensive caregiving platform that connects families with verified caregivers through AI-powered features, including health data analysis, activity tracking, and Natural Language Processing-driven note-taking for medical consultations.")

Section 3: The User (Who is this for?)

- Describe your target user in detail. Don't just say "students." Create a **User Persona**.
- **Example Persona:**
 - **Name:** Priya
 - **Age:** 45

- **Role:** Marketing Manager & mother of two.
- **Bio:** Priya lives in Bangalore while her elderly parents live alone in Pune. She worries constantly about their health and safety.
- **Needs:** A way to monitor her parents' well-being, manage their medical appointments, and find trusted help for them without having to call ten different agencies.
- **Frustrations:** "I feel guilty and overwhelmed. I don't know who to trust, and I can't be in two places at once."

The Mechanics (The "What" and "How")

This page details the features of your app and the technology that makes them smart.

Section 4: Key Features & User Flow

- Key Features: List the 3-5 main features of your app. For each feature, write one sentence explaining what it does.
 - **Example (for CareSync):**
- 1. **AI Caregiver Matching:** Matches families with vetted caregivers based on specific needs, location, and language.
- 2. **Shared Health Dashboard:** A simple, real-time dashboard for families to track vitals, medication schedules, and appointments.
- 3. **NLP Medical Notes:** Listens to a doctor's consultation and automatically generates a simple, summarized to-do list for the caregiver.
- 4. **Selfie-Based Activity Tracking:** Caregivers confirm tasks (like giving medication) with a simple, time-stamped selfie, giving families peace of mind.
- **Main User Flow:** Choose your most important feature and describe how a user would use it from start to finish, step-by-step.
 - Example (User Flow for "AI Caregiver Matching"):
- 1. User (Priya) opens the app and taps "Find a Caregiver."
- 2. She fills out a simple form (parent's needs, location, hours needed).
- 3. The AI shows her the top 3 matched caregivers with video intros and reviews.
- 4. Priya taps to book a virtual interview with her top choice.

Section 5: The "AI Inside" (Core Components)

- This is crucial. List what kind of AI your app needs to work. Connect each AI component to a feature you listed.
 - **Text Generation (LLM):** Used for the "NLP Medical Notes" to summarize doctor's orders.
 - **Computer Vision:** Used for the "Selfie-Based Activity Tracking" to verify the caregiver and the task.

- **Matching Algorithm (Machine Learning):** Used for the "AI Caregiver Matching" to find the best caregiver for Priya's needs.
- **Predictive Analytics:** (Optional) Could be used to analyze health data and flag potential issues before they become emergencies.

The Vision (The "Look" and "Inspiration")

This page shows what your app looks like and what inspired it.

Section 6: Interface Sketches (The Blueprint)

- On paper or with a simple digital tool, sketch the 2-3 most important screens of your app. You don't need to be an artist! Simple boxes and text are perfect.
- Screens to sketch:
 1. The Home Screen/Dashboard: What does the user see when they first open the app?
 2. Your Main Feature Screen: What does the screen for your "AI Waste Scanner" or "Career Finder" actually look like?
- Label the key buttons, text boxes, and output areas. This proves you have thought about the user's experience.

Section 7: Project Ideas & Inspiration

You can use one of these ideas or let them inspire your own unique concept.

- **AI Travel Planner:** A smart application to optimize urban commuting and tourism by leveraging AI for route planning, real-time traffic analysis, and personalized travel recommendations.
- **Elderly Education App:** A user-friendly platform tailored for senior citizens, offering accessible learning modules, social engagement tools, and cognitive skill development resources.
- **Trading Gym:** An immersive simulation platform for financial education, enabling users to practice trading strategies in a risk-free environment using AI-driven market insights.
- **Task Simplifier:** An AI-based tool to enhance productivity by automating and organizing daily tasks with intelligent prioritization and reminders.
- **CareSync (CareApp):** A comprehensive caregiving platform connecting families with verified caregivers through AI-powered features such as health data analysis, personalized dashboards, selfie-based two-factor authentication for activity tracking, and NLP-driven note-taking for medical consultations.
- **SecondChance App:** A holistic dashboard supporting career re-entry and skill enhancement, featuring AI-powered job matching, internship discovery, application tracking, mentorship connections, micro-loan facilities, and tailored training courses.
- **NeuroTrade:** An innovative AI-driven trading application addressing investor challenges (time constraints, inadequate research, emotional decision-making) by building

diversified portfolios, learning from market trends, and automating portfolio management per SEBI regulations.

- **Parking Ease App:** A solution to alleviate urban parking woes by using AI to locate available parking spaces in real time, reducing congestion and saving time.
- **Urban Mitra App:** A versatile application offering a suite of urban solutions, from service access to community engagement, powered by AI-driven recommendations.
- **InfraGuard:** An AI-enabled monitoring system to safeguard urban infrastructure by detecting anomalies, predicting maintenance needs, and ensuring operational efficiency.
- **Urban Heating Prevention Solution:** An application leveraging AI to propose actionable strategies for mitigating urban heat islands through data-driven environmental analysis.
- **Deal Spotter App:** A localized, AI-powered deal-finding platform that curates the best offers and discounts for users.

What to Submit: Your Final Capstone Blueprint

You will combine all sections into a single document (e.g., a PDF, Google Doc, or Word document) with the following 6 parts:

1. **Section 1: Project Name & The Problem**
2. **Section 2: The Solution (Your Mission)**
3. **Section 3: The User (Your User Persona)**
4. **Section 4: Key Features & User Flow**
5. **Section 5: The "AI Inside" (Core Components)**
6. **Section 6: Interface Sketches** (Embed photos/screenshots of your drawings)

Submission Guidelines

- Use AI tools like Chatgpt, Gemini or Perplexity to get help in case you get stuck in any section
- Compile your 6-section blueprint into a single document.
- Submit your completed document in the submission portal.
- This is your final project. Be proud of what you've designed!